

FREEMAN YIU

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EDUCATION

California State Polytechnic University, Pomona
Bachelor of Science, Computer Science; GPA: 3.94

Pomona, CA
Expected December 2026

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, JavaScript, R

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Transformers, Streamlit, FastAPI

Technologies: Git, GitHub, Docker, PostgreSQL, AWS, Excel, Cursor

EXPERIENCE

Cal Poly Pomona MEP-Wise

Pomona, CA

Computer Science Tutor

September 2025 - Present

- Mentor 20+ students in Math, Physics, and Computer Science by simplifying complex concepts through visual explanations, improving academic outcomes by 20-25%

Northrop Grumman Collaboration Project

Pomona, CA

Integration Software Engineer

August 2025 - Present

- Unify outputs from software, mechanical, and electrical subsystems while collaborating with a cross-functional team, enabling end-to-end autonomous vehicle operations
- Refactor and debug code in Python/C++ modules to strengthen system reliability and data flow reliability
- Deploy and monitor ROS 2 nodes in Ubuntu VMs, achieving real-time sensor communications between subsystems, and visualize component interactions through graphs to accelerate debugging
- Conduct GitHub code reviews with teammates to raise code quality and collaboration efficiency

Cal Poly Pomona Autonomous Vehicle Lab - UAV Project

Pomona, CA

Object-Detection Team Member

August 2025 - Present

- Engineer and evaluate PyTorch-based detection models to increase UAV perception accuracy in various situations
- Build data ingestion pipelines for image datasets, improving data quality and training efficiency
- Analyze loss curves and performance metrics to guide iterative tuning, ensuring model stability

Cal Poly Pomona CALSys Lab

Pomona, CA

Research Assistant

June 2025 - August 2025

- Developed Python ETL pipelines via TOR to retrieve cybersecurity-related data for downstream analytics
- Integrated a classification model to automate data labeling, minimizing manual workflow by 40%
- Leveraged PostgreSQL to query and store scraped data, enhancing data storage and retrieval by 30%

PROJECTS

Legal Document RAG System | *Python, LLM, Vector Databases, FastAPI*

- Architecting a RAG system using local Ollama LLM to answer questions over legal documents, reducing manual document lookup time
- Implementing PDF parsing, text chunking, embedding generation, and vector retrieval with ChromaDB to allow faster retrieval speed in context searching
- Developing backend FastAPI routes to expose retrieval functionality via REST APIs and support scalability

Diabetes Classifier | *Python, Pandas, Matplotlib, Scikit-learn*

- Analyzed and transformed ~250k medical records to create high-quality data for predictive modeling
- Conducted EDA and feature engineering to identify key predictors and extract insights
- Trained and evaluated multiple classifiers, achieving a ~15% improvement in predictive accuracy
- Architected an interactive Streamlit app enabling real-time predictions from user health inputs